REFINING A ZARISKI COVERING OF AN AFFINE SCHEME BY A STANDARD ZARISKI COVERING

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file topologies.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): the official pool candidate recorded statement [329, 338] and proof(s) [[340, 344]]; this spec covers the complete environment instead (statement 329-338, proof span(s) [(340, 344)]), since in this chapter the author's proof environment(s) follow the statement environment and some pool entries under-counted them.

Source reference. Original reference 01HT: sources/stacks-chow/tags.txt, source lines 1954–1954. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01HS: sources/stacks-chow/tags.txt, source lines 1953–1953. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let T be an affine scheme. Let $\{T_i \to T\}_{i \in I}$ be a Zariski covering of T. Then there exists a Zariski covering $\{U_j \to T\}_{j=1,\ldots,m}$ which is a refinement of $\{T_i \to T\}_{i \in I}$ such that each U_j is a standard open of T, see Schemes, Definition 01HT. Moreover, we may choose each U_j to be an open of one of the T_i.*

Proof. Follows as T is quasi-compact and standard opens form a basis for its topology. This is also proved in Schemes, Lemma 01HS. □

Date: stacks-project:topologies.tex:local-snapshot-20261012; GNU FDL 1.2 or later.